

A Speech Recognition model takes 80 million MAC operations to execute one inference

- (i) What is the operating frequency of a Cortex-M4 processor required to perform an inference within 500ms?
- (ii) Given that the processor (and its peripherals) consume 3mW per inference is powered by a battery with capacity of 3000 Whr (i.e. Joules in term of energy) , how long can the system operation?
- (iii) What are the requirements of the battery to be able to support the system operation over the full duration?

i) assuming, 1 MAC per clock cycle. no. of cycles = 80 000 000. Converting to frequency, $80000000/0.5 = 160\text{MHz}$

ii) 1 inference = 500ms, 3mW, energy per inference = $3\text{mW} * 0.5 = 1.5\text{mJ}$, total inference = $3000\text{J}/1.5\text{mJ} = 2000000$, which is 1000000 seconds ~ 11days.

iii) microcontrollers recently take 1.8v, 3.3v for peripherals, 5v for interface (eg. USB), Battery voltage must support operating voltage level required by system, Cortex-M operating voltage is 3.3v along with sufficient operating current across the whole duration. good thing about Li-ion battery, it maintains its operating voltage then it drops significantly only when reaching empty capacity, its a curve can be seen online. Normally, battery wont give very nice stable 3.3v, gives 3.7v, 1.5v etc. Thus usually power supply in a MCU is connected to a voltage regulator/converter, eg DC-DC switcher, buck-boost.